



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA1017 – SOFTWARE ENGINEERING

INDIVIDUAL REFLECTION REPORT

INTELLIGENT DISASTER EARLY WARNING AND EVACUATION MANAGEMENT SYSTEM

NAME: RAGUL SG

REGISTER NO.: 192525007

ROLE: Requirements Analysis and Problem Formulation

DESIGN / DEVELOPMENT DECISION: I helped convert the broad disaster-management problem into a structured set of system requirements for citizens, authorities, emergency responders, and administrators. I decided that the MVP should focus on an integrated decision-support workflow rather than attempting to implement every real-world disaster service.

CHALLENGE FACED AND SOLUTION: The main challenge was reducing a large and complex problem into realistic modules without losing important safety requirements. I addressed this by separating the platform into monitoring, risk assessment, alerts, GIS, evacuation, shelters, citizen reporting, and response coordination.

LEARNING GAINED: This work improved my understanding of requirements engineering, stakeholder analysis, functional and non-functional requirements, and the importance of defining clear project boundaries. I learned that a system intended for emergency use must communicate its limitations clearly and must not make unsupported claims about prediction accuracy or guaranteed safety.

SDG RELEVANCE: The project is connected to SDG 11 because it supports safer, more inclusive, and more resilient communities through coordinated disaster information and preparedness tools.

COURSE OUTCOME CONTRIBUTION: The work supports the course outcome related to requirements analysis and system design because I translated stakeholder needs into traceable requirements, modules, and evaluation criteria.

POSSIBLE IMPROVEMENT: In future, I would validate the requirements through interviews with emergency-management professionals and community representatives.